Mobile Phone Use Log Data Set

Description of data collection

This document describes the details of the study, in which the Mobile Phone Use Log Data Set was collected.

We believe that knowing the details of how the data was collected is important to correctly interpret it. The goal of this document is to provide the necessary background knowledge.

The text is a shortened version of the description of the study methodology found in the original scientific article:
Beyond Interruptibility: Predicting Opportune Moments to Engage Mobile Phone Users.
Martin Pielot, Bruno Cardoso, Kleomenis Katevas, Joan Serrà, Aleksandar Matic, Nuria Oliver. UbiComp '17: ACM International Joint Conference on Pervasive and Ubiquitous Computing, 2017.

Summary of Methodology

The goal of the original study was to explore whether patterns of phone use can predict the user's openness to engage with suggested content.

To systematically explore which type of information available in mobile phone applications indicates good moments to engage with users, we conducted a field study, which can be summarized as follows:

- For an average duration of 4 weeks, 337 participants installed a study application onto their primary mobile phones (Android only) and kept it running in the background.
- The application created about 10 to 15 notifications per day. Such notifications led to a mood questionnaire with 4 Likert-scale items that could be answered in less than 10 seconds. We collected 30,689 responses to the questionnaire.
- At the bottom of the questionnaire, two types of content were offered to the users, randomly chosen from a pool of eight different types of content, corresponding to different services (e.g., games, news, videos, etc.). In 3,367 cases, participants engaged with the offered content.
- While participants believed that the mood questionnaire was the main purpose of the study, what we were interested in was their engagement with the recommended content. Hence, we logged whether instead of simply closing the questionnaire, they opened one of the two suggested contents, which we refer to as engaging with the recommended content.
- At the same time, the application logged phone usage patterns from a wide range of sensors and other information sources.

Data Files

pinfo.csv

Details of the participants, such as internal ID, device type, age, Big 5 traits, average valence responses, …

Data

Folder which contains one file per participant. The filename matches the internal ID.
Each row represents one sensor event. Sensor events are sorted by the time. There are two types of fields (columns): fields that are present for every sensor and fields that are specific to one sensor.

Common fields are:

- **uuid**: the unique user ID, e.g. u73cbd0
- **version**: version number of the study application. This can be important to filter data, because some sensors were only introduced in later versions of the app
- **ts**: time stamp of the **local device time** when the sensor event occurred or was logged. Example: 6/23/2016 10:18:31
- **ts_raw**: unix time stamp, i.e. milliseconds since 1.1.1970 GMT (!), not local time!
- **sensor_id**: name of the sensor. Should match with the abbreviation in Sensors.xlsx

The values of each sensor are stored in fields that contain the sensor name + an additional identifier. For example, the acceleration sensor 'acc' creates two values:

- **Acc_Avg** = the average acceleration during the observation period
- **Acc_Max** = the maximum acceleration during the observation period

Sensors

To keep the overall energy consumption reasonable, we used three strategies to collect data:

1. System-wide broadcasted events, such as battery level, were collected always by registering to the respective broadcast receivers and callbacks.
2. Data from energy-intensive sensors, such as location, was collected every 10 minutes for 30 seconds.
3. Data related to the user's interaction with the phone, such as the app in foreground, was only collected while the screen was unlocked.

With these strategies, we limited the battery consumption to a level that the study application was shown in the OS internal battery usage view as consuming about the same amount of battery as other popular apps, such as Facebook or WhatsApp.

Sensor name	Logs when	Description
Acc	Every 10 min	Mean (Avg) and maximum (Max) acceleration of the phone in last 30 seconds
Airpln	On change	Airplane mode enabled
App	On change	Package name of app in foreground
Audio	On change	Whether e.g. headphones are used
Batt	Every 10 min	Battery level [0-100] in %

BattDrain	Every 10 min	Battery drain (%) in last 10 min
CellTower	Every 10 min	Info about the cell tower
Charging	Every 10 min	Whether phone is charging or not
Data	Every 10 min	Mean data usage in kb/min in the last 30 seconds
Esm	Completed questionnaire	Results of the experience sampling questionnaire
Light	Every 10 min	Mean ambient light level in SI lux units in the last 30 seconds
Noise	Every 10 min	Mean ambient noise level (dB) in the last 30 seconds
Notif	On event	Fired when notification is posted
Notif	On event	Fired when notification is removed or posted
NotifCenter	On event	Fired when user opens notification center, values contains the number of pending notifications
NotifDismissed	On event	Fired when user dismisses a single notification
NotifsCleared	On event	Fired when user dismisses all notifications
Orient	Every 10 min	Device orientation
Phone	On event	Fired on incoming or outgoing phone call
Photo	Every 10 min	Photo name and date
Prox	On event	Whether screen is covered (1) or not (0)
Screen	On change	Fired when screen is turned on, off, or unlocked
ScrOrient	Every 10 min	Whether the screen changed oriantation (e.g. Landscape or Portrait)
Loc	Every 10 min	Whether user is at home, work, a significant place, or somewhere else
Sms	On event	Sms Events: sent, received, seen
UserAct	On change	User activity as inferred by Google API
Wifi	Every 10 min	Details on Wifi connection

In the following, we describe the details of each sensor

Acc (Acceleration)

Work-in-progress